INCH-POUND

AN839 Rev 9
w/AMENDMENT 1
18 November 2024
SUPERSEDING
AN839 Rev 9
1 October 2014

DETAIL SPECIFICATION SHEET

ELBOW, TUBE TO HOSE, 45 DEGREE

Reinstated after 7 June 2011. Inactive for new design.
For new design, use SAE-AS5182.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and SAE-AS4843/1.

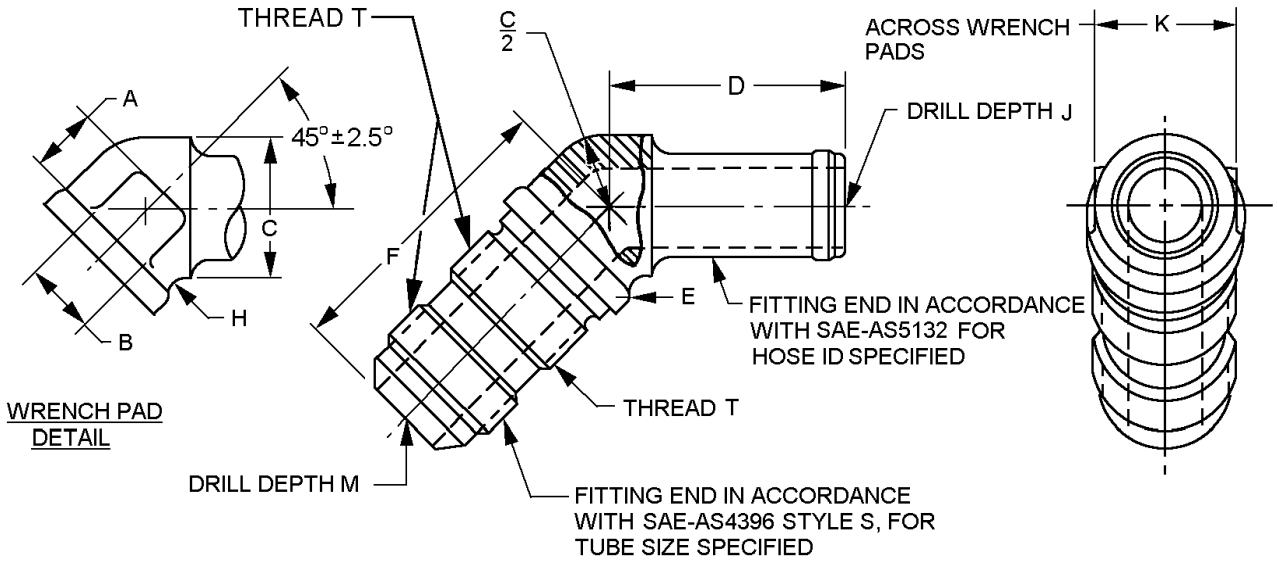


FIGURE 1. 45°Elbow dimensions and configurations.

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Dash number	Hose ID inches (mm)	Tube OD inches (mm)	Thread T SAE-AS8879	A Approx inches (mm)	B Approx inches (mm)
4	.250 (6.35)	.250 (6.35)	.4375-20UNJF-3A	.219 (5.56)	.250 (6.35)
6	.375 (9.53)	.375 (9.53)	.5625-18UNJF-3A	.281 (7.14)	.375 (9.53)
8	.500 (12.70)	.500 (12.70)	.7500-16UNJF-3A	.344 (8.74)	.438 (11.13)
10	.625 (15.88)	.625 (15.88)	.8750-14UNJF-3A	.438 (11.13)	.500 (12.70)
12	.750 (19.05)	.750 (19.05)	1.0625 -12UNJF-3A	.500 (12.70)	.566 (14.38)
16	1.000 (25.40)	1.000 (25.40)	1.3125 -12UNJ-3A	.563 (14.30)	.688 (17.48)
20	1.250 (31.75)	1.250 (31.75)	1.6250 -12UNJF-3A	.688 (17.48)	1.000 (25.40)
24	1.500 (38.10)	1.500 (38.10)	1.8750 -12UNJF-3A	.750 (19.05)	1.125 (28.58)

Dash number	C dia. inches (mm)	D +.047 (1.19) -.000 inches (mm)	E Radius inches (mm)	F +.047 (1.19) -.000 inches (mm)
4	.438 (11.13)	1.688 (42.88)	.063 (1.60)	1.422 (36.12)
6	.566 (14.38)	1.719 (43.66)		1.563 (39.70)
8	.750 (19.05)	1.750 (44.45)	.094 (2.39)	1.828 (46.43)
10	.875 (22.23)	1.797 (45.64)		2.047 (51.99)
12	1.063 (27.00)	1.828 (46.43)		2.281 (57.94)
16	1.313 (33.35)	1.890 (48.01)	.125 (3.18)	2.391 (60.73)
20	1.625 (41.28)	1.969 (50.01)		2.547 (64.69)
24	1.875 (47.63)	2.031 (51.59)		2.672 (67.87)

Dash number	H Radius inches (mm)	J +.047 (1.19) -.000 inches (mm)	K inches (mm)	M +.047 (1.19) -.000 inches (mm)
4	.063 (1.60)	1.750 (44.45)	.438 (11.13)	1.468 (37.29)
6	.094 (2.39)	1.797 (45.64)	.563 (14.30)	1.609 (40.87)
8		1.813 (46.05)	.750 (19.05)	1.938 (49.23)
10	.125 (3.18)	1.875 (47.63)	.875 (22.23)	2.172 (55.17)
12		1.953 (49.61)	1.063 (27.00)	2.422 (61.52)
16		2.047 (51.99)	1.313 (33.35)	2.593 (65.86)
20	.188 (4.78)	2.172 (55.17)	1.625 (41.28)	2.797 (71.04)
24		2.281 (57.94)	1.875 (47.63)	2.969 (75.41)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances are ±.016 inch (0.41 mm).
4. Break sharp edges and remove all hanging burrs and slivers
5. Machined surfaces shall be finished to 125µ-in Ra, forged surfaces shall be 250µ-in Ra, unless otherwise specified on the figures. Surface finish shall be in accordance with ASME B46.1.
6. For design features purposes, this standard takes precedence over documents referenced herein.

FIGURE 1. 45° Elbow dimensions and configurations - Continued.

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REQUIREMENTS:

Dimensions and configuration shall be in accordance with figure 1.

Installation shall be in accordance with MS21344.

Materials and finishes: Unless otherwise specified materials and finishes shall be in accordance with SAE-AS4843; see table I for material/finish code.

TABLE I. Material and finish code letters.

Material and finish code	Material	Protective chemical finish 3/
B 1/	Copper alloy, type 377 forging in accordance with ASTM B124/B124M or half hard forging or bar in accordance with ASTM B138/B138M or bar in accordance with SAE-AMS4614.	No finish.
BC 1/	Copper alloy, type 377 forging in accordance with ASTM B124/B124M or half hard forging or bar in accordance with ASTM B138/B138M or bar in accordance with SAE-AMS4614.	Cadmium in accordance with SAE-AMS-QQ-P-416, type II, class 3.
D 2/	Type 2014-T6 aluminum alloy forging in accordance with SAE-AMS-QQ-A-367 or SAE-AMS4133, or type 2024-T6 aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/6, or type 2024-T851 aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/6 or SAE-AMS4339.	Anodize in accordance with SAE-AMS2472 or MIL-PRF-8625, type II, class 2, dye blue, duplex seal in accordance with procurement specification.
DV 2/	Type 2014-T6 aluminum alloy forging in accordance with SAE-AMS-QQ-A-367 or SAE-AMS4133, or type 2024-T6 aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/6, or type 2024-T851 aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/6 or SAE-AMS4339.	High purity aluminum in accordance with MIL-DTL-83488, class 3, type II with maximum coating thickness of .0005 inch. Glass bead peen pressure shall be 25 psi (1.72 bar) maximum.
J	Type 304 corrosion resistant steel forging or bar in accordance with SAE-AMS-QQ-S-763 or SAE-AMS5639.	Passivate in accordance with SAE-AMS2700, type VI or VII.
K	Type 316 corrosion resistant steel forging or bar in accordance with SAE-AMS-QQ-S-763 or SAE-AMS5648.	Passivate in accordance with SAE-AMS2700, type VI or VII.

See notes at end of table

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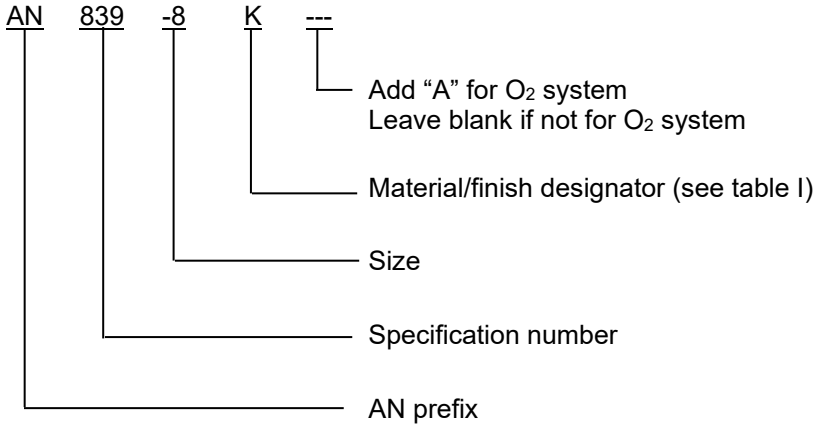
TABLE I. Material and finish code letters - Continued.

Material code	Material	Protective chemical finish 3/
R	Type 321 corrosion resistant steel forging or bar in accordance with SAE-AMS-QQ-S-763 or SAE-AMS5645.	Passivate in accordance with SAE-AMS2700, type VI or VII.
W 2/	Type 7075-T73 aluminum alloy forging in accordance with SAE-AMS-QQ-A-367 or SAE-AMS4141, or type 7075-T73 aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/9, in accordance with type 7075-T7351 Aluminum alloy bar in accordance with SAE-AMS4124.	Anodize in accordance with SAE-AMS2472 or MIL-PRF-8625, type II, class 2, dye brown similar to color 10080 in accordance with SAE AMS-STD-595, duplex seal in accordance with procurement specification.
WV 2/	Type 7075-T73 aluminum alloy forging in accordance with SAE-AMS-QQ-A-367 or SAE-AMS4141, or type 7075-T73 aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/9, in accordance with type 7075-T7351 Aluminum alloy bar in accordance with SAE-AMS4124.	High purity aluminum in accordance with MIL-DTL-83488, class 3, type II with maximum coating thickness of .0005 inch. Glass bead peen pressure shall be 25 psi (1.72 bar) maximum.

1/ Material code was dash on previous revisions, changed to agree with SAE-ARP1590.
2/ Aluminum code D is inactivated, use code W.
3/ Cadmium shall not be used in potable water systems.

Oxygen systems for aerospace, shipboard, and ground support equipment. Parts for use in oxygen systems (O_2) shall be identified in the PIN as code “A” and shall be furnished cleaned, packaged, and labeled in accordance with SAE-AS611 to a process approved by the user.

Part or Identifying Number (PIN): The PIN consists of the prefix “AN”, specification sheet number, dash number for hose and flared tube size, material finish code letter(s) and blank or “A” for oxygen cleaning requirements. Unassigned PIN’s shall not be used.



PIN example: AN839-8K indicates a 45° elbow, tube to hose, .500 inches (12.70 mm), CRES alloy.

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Marking: Part shall be permanently marked with the AN PIN, and include the manufacturers CAGE, name, or trademark.

Guidance on use of alternative parts with less hazardous or non-hazardous materials. This specification provides for a number of alternative plating materials via the PIN. Users should select the PIN with the least hazardous material that meets the form, fit, and function requirements of their application.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

Canceled AN PIN and superseding AN PIN see table II.

TABLE II. Superseding AN PIN.

Canceled AN PIN	Hose Size	Pipe Size	Superseding AN PIN Inactive part
AN839-4	0.250	0.125	AN839-4B
AN839-6	0.375	0.250	AN839-6B
AN839-8	0.500	0.375	AN839-8B
AN839-10	0.625	0.500	AN839-10B
AN839-12	0.750	0.750	AN839-12B
AN839-16	1.000	0.750	AN839-16B
AN839-20	1.250	1.250	AN839-20B
AN839-24	1.500	1.500	AN839-24B
AN839-4S	0.250	0.125	AN839-4R
AN839-6S	0.375	0.250	AN839-6R
AN839-8S	0.500	0.375	AN839-8R
AN839-10S	0.625	0.500	AN839-10R
AN839-12S	0.750	0.750	AN839-12R
AN839-16S	1.000	0.750	AN839-16R
AN839-20S	1.250	1.250	AN839-20R
AN839-24S	1.500	1.500	AN839-24R

Table III provides a detailed cross-reference of AN839 PINs and replacement SAE-AS5182 PINs. Users are cautioned to evaluate replacements for their particular application.

Metal cracking due to high temperatures CRES alloy 347 “S” designator has been replaced by CRES alloy 321 “R” designator.

CAUTION: The superseding information is valid as of the date of this specification and may be superseded by subsequent revisions of the superseding document.

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TABLE III. Cross-reference data. 1/

AN PIN Inactive part	Hose Size	Pipe Size	AS PIN Active part AS PIN
AN839-4B	0.250	0.250	AS5182B04
AN839-4BC	0.250	0.250	None
AN839-4D	0.250	0.250	AS5182W04
AN839-4J	0.250	0.250	AS5182J04
AN839-4K	0.250	0.250	AS5182K04
AN839-4R	0.250	0.250	AS5182R04
AN839-4S	0.250	0.250	AS5182R04
AN839-4W	0.250	0.250	AS5182W04
AN839-4WV	0.250	0.250	AS5182WV04
AN839-6B	0.375	0.375	AS5182B06
AN839-6BC	0.375	0.375	None
AN839-6D	0.375	0.375	AS5182W06
AN839-6J	0.375	0.375	AS5182J06
AN839-6K	0.375	0.375	AS5182K06
AN839-6R	0.375	0.375	AS5182R06
AN839-6S	0.375	0.375	AS5182R06
AN839-6W	0.375	0.375	AS5182W06
AN839-6WV	0.375	0.375	AS5182WV06
AN839-8B	0.500	0.500	AS5182B08
AN839-8BC	0.500	0.500	None
AN839-8D	0.500	0.500	AS5182W08
AN839-8J	0.500	0.500	AS5182J08
AN839-8K	0.500	0.500	AS5182K08
AN839-8R	0.500	0.500	AS5182R08
AN839-8S	0.500	0.500	AS5182R08
AN839-8W	0.500	0.500	AS5182W08
AN839-8WV	0.500	0.500	AS5182WV08
AN839-10B	0.625	0.625	AS5182B10
AN839-10BC	0.625	0.625	AS5182B10
AN839-10D	0.625	0.625	AS5182W10
AN839-10J	0.625	0.625	AS5182J10
AN839-10K	0.625	0.625	AS5182K10
AN839-10R	0.625	0.625	AS5182R10
AN839-10S	0.625	0.625	AS5182R10
AN839-10W	0.625	0.625	AS5182W10
AN839-10WV	0.625	0.625	AS5182WV10

See notes at end of table.

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TABLE II. Cross-reference data - Continued. 1/ 2/

AN PIN Inactive part	Hose Size	Pipe Size	AS PIN Active part
AN839-12	0.750	0.750	AS5182B12
AN839-12BC	0.750	0.750	None
AN839-12D	0.750	0.750	AS5182W12
AN839-12J	0.750	0.750	AS5182J12
AN839-12K	0.750	0.750	AS5182K12
AN839-12R	0.750	0.750	AS5182R12
AN839-12S	0.750	0.750	AS5182R12
AN839-12W	0.750	0.750	AS5182W12
AN839-12Wv	0.750	0.750	AS5182Wv12
AN839-16B	1.000	1.000	AS5182B16
AN839-16BC	1.000	1.000	None
AN839-16D	1.000	1.000	AS5182W16
AN839-16J	1.000	1.000	AS5182J16
AN839-16K	1.000	1.000	AS5182K16
AN839-16R	1.000	1.000	AS5182R16
AN839-16S	1.000	1.000	AS5182R16
AN839-16W	1.000	1.000	AS5182W16
AN839-16WV	1.000	1.000	AS5182WV16
AN839-20B	1.250	1.250	AS5182B20
AN839-20BC	1.250	1.250	None
AN839-20D	1.250	1.250	AS5182W16
AN839-20J	1.250	1.250	AS5182J20
AN839-20K	1.250	1.250	AS5182K20
AN839-20R	1.250	1.250	AS5182R20
AN839-20S	1.250	1.250	AS5182R20
AN839-20W	1.250	1.250	AS5181W20
AN839-20WV	1.250	1.250	AS5181WV20
AN839-24B	1.500	1.500	AS5182B24
AN839-24BC	1.500	1.500	None
AN839-24D	1.500	1.500	AS5182W24
AN839-24J	1.500	1.500	AS5182J24
AN839-24K	1.500	1.500	AS5182K24
AN839-24R	1.500	1.500	AS5182R24
AN839-24S	1.500	1.500	AS5182R24
AN839-24W	1.500	1.500	AS5182W24
AN839-24WV	1.500	1.500	AS5182WV24

1/ For new design use material designators R and W.
2/ SAE “B” designator for copper alloy parts can be either no finish or cadmium plate.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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Referenced documents. In addition to SAE-AS4843/1, this document references the following:

SAE AMS-STD-595	SAE-AMS-QQ-P-416	SAE-AMS5645
MIL-PRF-8625	SAE-AMS-QQ-S-763	SAE-AMS5648
MIL-DTL-83488	SAE-AMS2472	SAE-ARP1590
MS21344	SAE-AMS2700	SAE-AS611
ASME B46.1	SAE-AMS4124	SAE-AS4396
ASTM B124/B124M	SAE-AMS4133	SAE-AS4843
ASTM B138/B138M	SAE-AMS4141	SAE-AS5132
SAE-AMS-QQ-A-225/6	SAE-AMS4339	SAE-AS5182
SAE-AMS-QQ-A-225/9	SAE-AMS4614	SAE-AS8879
SAE-AMS-QQ-A-367	SAE-AMS5639	

CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force – 71
DLA - CC

Review activities:
Navy - SA

Preparing activity:
DLA - CC

(Project 4730-2025-001)

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